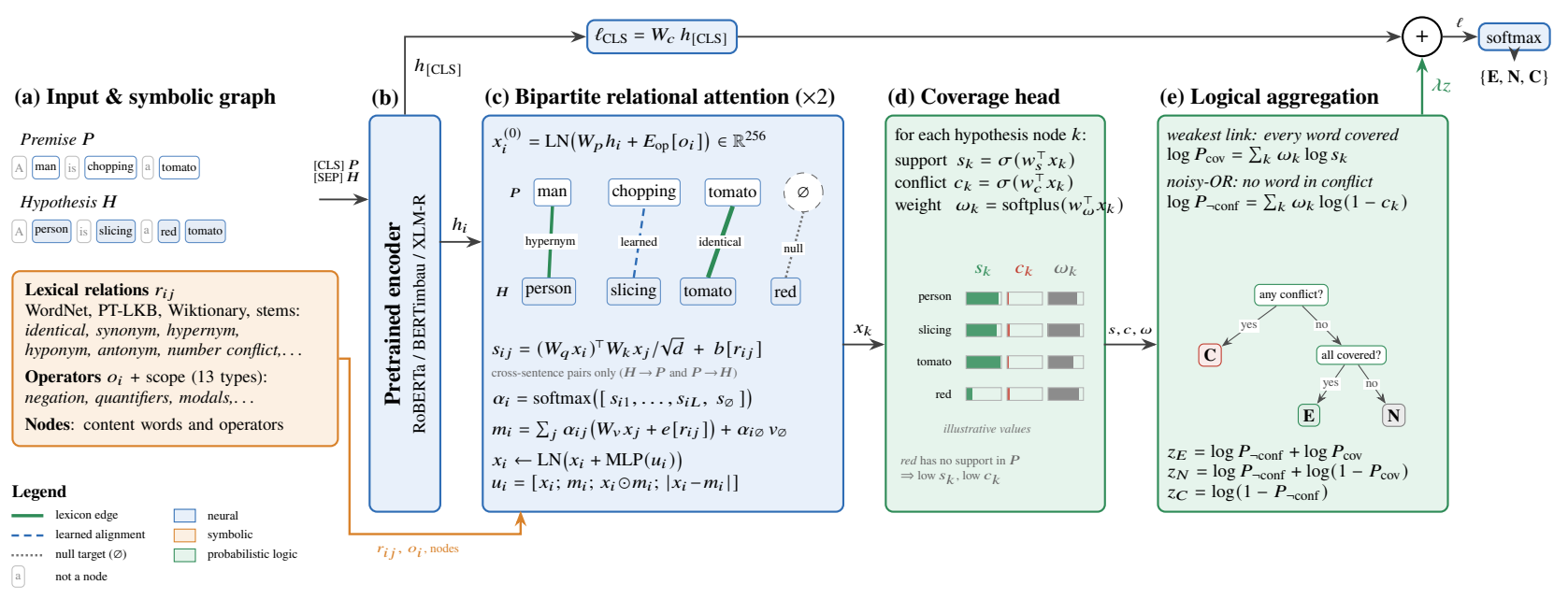

(a) Input & symbolic graph
Premise P
A man is chopping a tomato
Hypothesis H
A person is slicing a red tomato
Lexical relations r_{ij}
WordNet, PT-LKB, Wiktionary, stems:
identical, synonym, hypernym,
hyponym, antonym, number conflict,...
Operators o_i + scope (13 types):
negation, quantifiers, modals,...
Nodes: content words and operators
Legend
lexicon edge
learned alignment
null target (∅)
not a node
neural
symbolic
probabilistic logic
[CLS] P
[SEP] H
(b)
Pretrained encoder
RoBERTa / BERTimbau / XLM-R
h_i
$h_{[CLS]}$
r_{ij}, o_i, nodes
(c) Bipartite relational attention (×2)
$x_i^{(0)} = \mathrm{LN}(W_p h_i + E_{\mathrm{op}}[o_i]) \in \mathbb{R}^{256}$
P man chopping tomato ∅
hypernym learned identical null
H person slicing tomato red
$s_{ij} = (W_q x_i)^\top W_k x_j / \sqrt{d} + b[r_{ij}]$
cross-sentence pairs only ($H \to P$ and $P \to H$)
$\alpha_i = \mathrm{softmax}([\, s_{i1}, \ldots, s_{iL},\ s_{i\varnothing} \,])$
$m_i = \sum_j \alpha_{ij} (W_v x_j + e[r_{ij}]) + \alpha_{i\varnothing} v_\varnothing$
$x_i \leftarrow \mathrm{LN}(x_i + \mathrm{MLP}(u_i))$
$u_i = [x_i;\ m_i;\ x_i \odot m_i;\ |x_i - m_i|]$
x_k
(d) Coverage head
for each hypothesis node k:
support $s_k = \sigma(w_s^\top x_k)$
conflict $c_k = \sigma(w_c^\top x_k)$
weight $\omega_k = \mathrm{softplus}(w_\omega^\top x_k)$
s_k c_k ω_k
person
slicing
tomato
red
illustrative values
red has no support in P
⇒ low s_k, low c_k
s, c, ω
(e) Logical aggregation
weakest link: every word covered
$\log P_{\mathrm{cov}} = \sum_k \omega_k \log s_k$
noisy-OR: no word in conflict
$\log P_{\neg\mathrm{conf}} = \sum_k \omega_k \log(1 - c_k)$
any conflict?
yes
no
C
all covered?
yes
no
E
N
$z_E = \log P_{\neg\mathrm{conf}} + \log P_{\mathrm{cov}}$
$z_N = \log P_{\neg\mathrm{conf}} + \log(1 - P_{\mathrm{cov}})$
$z_C = \log(1 - P_{\neg\mathrm{conf}})$
$\ell_{\mathrm{CLS}} = W_c\, h_{[CLS]}$
+
ℓ
softmax
{E, N, C}
λz